

Curriculum Vitae

BEN WEBSTER

Address: Department of Pure Mathematics Email: ben.webster@uwaterloo.ca
University of Waterloo Website: <https://bwebste.github.io/>
Waterloo, ON, Canada Citizenships: Canadian, US

Employment/Education:

2017 – **Associate Professor**, University of Waterloo.
2017 – **Associate Faculty**, Perimeter Institute.
2016 – 2017 **Associate Professor**, University of Virginia.
2013 – 2016 **Assistant Professor**, University of Virginia.
2011 – 2013 **Assistant Professor**, Northeastern University.
2010 – 2011 **Assistant Professor**, University of Oregon.
2008 – 2010 **C.L.E. Moore Instructor and NSF Postdoctoral Fellow**, M.I.T.
 Sponsoring Scientist: R. Bezrukavnikov.
2007 – 2008 **Member and NSF Postdoctoral Fellow**, Institute for Advanced Study.
2002 – 2007 **Ph.D. in Mathematics**, University of California, Berkeley.
 Supervisor: N. Reshetikhin.
 Thesis: “Algebraic Poisson Geometry in Representation Theory and Combinatorics.”
1998 – 2002 **B.A. in Mathematics**, Simon’s Rock College.
 Supervisor: W. Dunbar

As visitor

2014 Spring **Junior Chair**, Université Denis Diderot–Paris VII,
 sponsored by Fondation Sciences Mathématiques de Paris.
2006 Fall Center for the Topology and Quantization of Moduli Spaces (Århus, Denmark).
2001 Spring Budapest Semesters in Mathematics (Budapest, Hungary).

Scientific/Academic Honors and Grants:

2024 – 2029 NSERC Discovery Grant (\$44k per year)
2023 SMRI International Visitor Program (\$5k for research visit to Australia)
2019 Golden Jubilee Research Excellence Award (UW Faculty of Math)
2018 – 2024 NSERC Discovery Grant (\$28k per year)
2016 Cory Family Teaching Award (recognizes excellence in lower division teaching;
 one awarded for all STEM fields at UVA per year)
2015 Kavli Fellow, National Academy of Sciences
2015 International Researcher Collaboration Award from Sydney University
2014 – 2016 Sloan Research Fellowship (\$50,000)
2012 – 2017 NSF CAREER grant: “Representation theory of symplectic singularities” (\$416,905)
2013 NSF conference grant “Algebra, Combinatorics, and Representation Theory.” (\$40,000)
2010 – 2012 NSA Young Investigator Grant (\$30,000)
2007 – 2011 NSF Postdoctoral Research Fellowship
2007 June Clay Liftoff Fellowship
2003 – 2007 NSF Graduate Research Fellowship

Research Interests:

Connections of representation theory and topology with quantum field theory.

Selected Publications:

available at https://arxiv.org/a/webster_b_1.html

full list of 59 publications and 13 preprints in Publication List

- 2025 1. *Twisted traces on abelian quantum Higgs and Coulomb branches*. (with Davide Gaiotto, Justin Hilburn, Jaime Redondo-Yuste, and Zheng Zhou) [arXiv:2307.10406](#), to appear in *Communications in Mathematical Physics*
2. *Bubbles in the affine Brauer and Kauffman categories*. (with Alistair Savage) [arXiv:2408.07000](#), to appear in *Journal of the Australian Mathematical Society*
3. *Homological mirror symmetry for hypertoric varieties II*. (with Benjamin Gammage and Michael McBreen). [arXiv:1903.07928](#); to appear in *Geometry and Topology*
4. *On the representation theory of Schur algebras in type B*. (with Dinushi Munasinghe) *J. Algebra* **667** (2025), 211–258. [arXiv:2307.10406](#)
- 2024 5. *Gelfand-Tsetlin modules: canonicity and calculations* (with Turner Silverthorne). *Algebras and Representation Theory* **27**, no. 2, 1405–1455. [arXiv:2011.06029](#)
6. *Gelfand-Tsetlin modules in the Coulomb context*. [arXiv:1904.05415](#); *Annals of Representation Theory* **1** (2024), no. 3, 393–437.
7. *RoCK blocks for affine categorical representations*. to appear in *Journal of Combinatorial Algebra*. [arXiv:2301.01613](#)
8. *Homological mirror symmetry for hypertoric varieties I*. (with Michael McBreen). *Geometry and Topology*. **28** (2024), no. 3, 1005–1063 [arXiv:1804.10646](#)
9. *Lie algebra actions on module categories for truncated shifted Yangians*. (with J. Kamnitzer, A. Weekes and O. Yacobi). *Forum of Mathematics, Sigma* **12** (2024). [arXiv:2203.12429](#)
10. *The nil-Brauer category*. (with Jon Brundan and Weiqiang Wang) *Annals of Representation Theory*, Volume 1 (2024) no. 1, pp. 21–58. [arXiv:2305.03876](#)
- 2023 11. *The degenerate Heisenberg category and its Grothendieck ring*. (with Jonathan Brundan and Alistair Savage). *Annales Scientifiques de l'École Normale Supérieure* **56** (2023) no. 5, pp. 1517–1563. [arXiv:1812.03255](#)
12. *Rational Cherednik algebras of $G(\ell, p, n)$ from the Coulomb perspective* (with Elise LePage) *Advances in Mathematics* **433** (2023), Paper No. 109295, 49 pp. [arXiv:1912.00046](#)
13. *3-dimensional mirror symmetry* (with Philsang Yoo). *Notices of the American Mathematical Society* **70** (2023), no. 9, 1395–1406 [arXiv:2308.06191](#)
- 2020 14. *Heisenberg and Kac-Moody categorification* (with Jonathan Brundan and Alistair Savage). *Selecta Math. (N.S.)* **26** (2020), no. 5, 74. [arXiv:1907.11988](#)
15. *On the definition of quantum Heisenberg category*. (with Jonathan Brundan and Alistair Savage). *Algebra and Number Theory* **14** (2020), no. 2, 275–321. [arXiv:1812.04779](#)
16. *A quantum Mirković–Vybornov isomorphism*. (with A. Weekes and O. Yacobi). *Representation Theory* **24** (2020), 38–84. [arXiv:1706.03841](#)
- 2019 17. *Weighted Khovanov-Lauda-Rouquier algebras*. to appear in *Documenta Mathematica*. *Doc. Math.* **24** (2019), 209250. [arXiv:1209.2463](#)
18. *On category $\mathcal{O}$ for affine Grassmannian slices and categorified tensor products*. (with J. Kamnitzer, P. Tingley, A. Weekes and O. Yacobi). *Proc. Lond. Math. Soc.* (3) **119** (2019), no. 5, 11791233. [arXiv:1806.07519](#)
- 2018 19. *Categorified skew Howe duality and comparison of knot homologies* (with Marco Mackaay). *Advances in Math* **330** (2018), 876–945. [arXiv:1502.06011](#)
- 2017 20. *Rouquier’s conjecture and diagrammatic algebra*. *Forum of Mathematics Sigma* **5** (2017), e27, 71 pp. [arXiv:1306.0074](#)

21. *A geometric construction of colored HOMFLYPT homology* (with G. Williamson). *Geometry & Topology* **21-5** (2017), 2557–2600. [arXiv:0905.0486](#)
22. *On generalized category $\mathcal{O}$ for a quiver variety*. *Mathematische Annalen* **368** (2017), no. 1–2, 483–536. [arXiv:1409.4461](#)
23. *Knot invariants and higher representation theory*. *Memoirs of the American Mathematical Society* **250**, no. 1191, pp. 133. [arXiv:1309.3796](#)
- 2016 24. *Quantizations of conical symplectic resolutions II: category $\mathcal{O}$ and symplectic duality*. (with T. Braden, A. Licata and N. Proudfoot). *Astérisque* No. 384, 75–179. [arXiv:1407.0964](#)
25. *Quantizations of conical symplectic resolutions I: local and global structure* (with T. Braden and N. Proudfoot). *Astérisque* No. 384, 1–73. [arXiv:1208.3863](#)
26. *Mirković-Vilonen polytopes and Khovanov-Lauda-Rouquier algebras* (with Peter Tingley). *Compositio Mathematica* **152**, no. 8, 1648–1696. [arXiv:1210.6921](#)
27. *Tensor product categorifications and the super Kazhdan-Lusztig conjecture* (with Jonathan Brundan and Ivan Losev). *International Mathematics Research Notices*, Volume 2017, Issue 20, Pages 6329–6410. [arXiv:1310.0349](#)
- 2015 28. *On uniqueness of tensor products of irreducible categorifications* (with Ivan Losev). *Selecta Math. (N.S.)* **21**, no. 2, 345–377. [arXiv:1303.1336](#)
29. *Canonical bases and higher representation theory*. *Compositio Mathematica* **151**, no. 1, 121–166. [arXiv:1209.0051](#)
- 2014 30. *Yangians and quantizations of slices in the affine Grassmannian* (with J. Kamnitzer, A. Weekes and O. Yacobi). *Journal of Algebra and Number Theory* **8** (2014), no. 4, 857–893. [arXiv:1209.0349](#)
- 2012 31. *Hypertoric category $\mathcal{O}$* (with T. Braden, A. Licata, and N. Proudfoot). *Advances in Mathematics*. **231** (2012), no. 3-4, 1487–1545. [arXiv:1010.2001](#)
- 2011 32. *The geometry of Markov traces* (with G. Williamson). *Duke Mathematics Journal*, **160** (2011) 401–419. [arXiv:0911.4494](#)
- 2010 33. *Gale duality and Koszul duality* (with T. Braden, A. Licata, and N. Proudfoot). *Advances in Mathematics*, **225** (2010) 2002–2049. [arXiv:0806.3256](#)
- 2008 34. *A geometric model for the Hochschild homology of Soergel bimodules* (with G. Williamson). *Geometry and Topology*, **12** (2008) 1243–1263. [arXiv:0707.2003](#).

Students & postdocs supervised:

- | | |
|-------------|---|
| 2016 – 2019 | Chris Leonard (PhD, Virginia; joint with Weiqiang Wang) |
| 2017 – 2019 | Alex Weekes (postdoc, Perimeter) |
| 2017 – 2020 | Qiao Zhou (postdoc, Perimeter) |
| 2018 – | Aiden Sutter (MSc, PSI; PhD, Perimeter/Waterloo) |
| 2019 | Amanda Garcia (MMath, Waterloo) |
| 2019 | Edward Poon (MMath, Waterloo) |
| 2019 | Elise LePage (MSc, PSI) |
| 2019 – 2020 | Turner Silverthorne (undergrad, Waterloo) |
| 2019 | Jerry Guan (undergrad, Waterloo) |
| 2019 – | Justin Hilburn (postdoc, Perimeter) |
| 2019 – 2020 | Mark Penney (postdoc, Perimeter) |
| 2020 – 2022 | Dene Lepine (PhD, Waterloo) |
| 2020 – | Dinushi Munasinghe (PhD, Toronto) |
| 2020 | Baorui Zhou (undergrad, Waterloo) |

2021	Jeremy Peters (MSc, PSI)
2021	Shannon Jeffries (undergrad, Waterloo)
2021 – 2023	Ethan Kowalenko (postdoc, Waterloo)
2021 –	Keke Zhang (PhD, Perimeter/Waterloo)
2023	Zev Friedman (undergrad, Waterloo)
2023 –	Nikita Grygoryev (PhD, co-supervision with Gaiotto and Costello, Perimeter/Waterloo)
2024	Ramana Murugesan (MMath)
2024 –	Meri Zaimi (postdoc, Perimeter)
2025 –	Jack Jia (PhD, Waterloo)
2025 –	Kledion Shahidi (PhD, co-supervision with Michael Borinsky, Waterloo)

Selected Lectures (since 2020):

2025	Oct.	Perimeter Institute (Mathematical Physics Seminar): <i>The geometry of Coulomb branches</i>
	Jul.	U. of York (Geometric and categorical Lie theory): <i>KLRW algebras under pressure</i>
	Jun.	BIMSA Beijing (String-Math): <i>The geometry of Coulomb branches</i>
	Jun.	SNU Seoul : <i>Non-degeneracy of Heisenberg and Kac-Moody categorifications</i>
	Jun.	CUHK (MIST workshop): <i>Geometry of Coulomb branches</i>
	May	Sustech (International Conference on Representations of Lie Algebras): <i>Gelfand-Tsetlin modules and KLRW algebras</i>
	Apr.	Yale (Geometry and Symmetry Seminar): <i>Geometry of Coulomb branches</i>
	Apr.	Harvard CMSA (Colloquium): <i>3d mirror symmetry</i>
	Apr.	Stony Brook (Symplectic Singularities, Supersymmetric QFT, and Geometric Representation Theory): <i>Geometry of Coulomb branches</i>
	Feb.	Toronto (Mathematical Physics Seminar): <i>Geometry of Coulomb branches</i>
	Jan.	Zoom (Imperial Mathematical Physics Seminar): <i>Gluing constructions for Coulomb branches</i>
2024	Oct.	Virginia (Algebra Seminar): <i>Coulomb branches and representation theory</i>
	Oct.	Virginia (Colloquium): <i>Representation theory and a little bit of quantum field theory</i>
	Oct.	NCSU (Quantum Groups and Representation Theory): <i>Coulomb branches and representation theory</i>
	Sep.	Les Diablerets (Quantization in Representation Theory, Derived Algebraic Geometry, and Gauge Theory): <i>3d mirror symmetry</i>
	Apr.	Berkeley (String-Math Seminar): <i>The cotangent bundle of the Grassmannian as a Coulomb branch</i>
	Mar.	Zoom (Quivers meeting): <i>Twisted traces and symplectic singularities</i>
	Jan.	Ottawa (Algebra Seminar): <i>Gelfand-Tsetlin modules and other fun with Coulomb branches</i>
	Jan.	Ottawa (Colloquium): <i>Representation theory and a little bit of quantum field theory</i>
2023	Sep.	Pollica (Supersymmetric Quantum Field Theory and Mathematics): <i>3d mirror symmetry</i>
	Aug.	Edinburgh (Geometric Representation Theory and W-algebras): <i>Finite W-algebras as Coulomb branches</i>
	Jun.	MIT (Summer School in Geometric Representation Theory): lead speaker at 1 week summer school
	Mar.	Imperial (Mathematical Physics Seminar): <i>Noncommutative resolutions and Coulomb branches</i>
	Feb.	Sydney (Categorification in representation theory): <i>Categorical $\mathfrak{sl}_2$ actions and RoCK blocks</i> <i>Categorical $\mathfrak{sl}_2$ representations and modular representation theory</i> (expository 4 lecture series)
	Jan.	Melbourne (Representation Theory Seminar): <i>Noncommutative resolutions and Coulomb branches</i>

- Jan. **Singapore** (Workshop on interactions between representation theory, combinatorics, and geometry): *The noncommutative Springer resolution of type A and KLRW algebras*
- 2022 Dec. **AIM** (Floer theory of symmetric products and Hilbert schemes):
Fun with Coulomb branches
- Sep. **Oberwolfach** (Character Theory and Categorification):
Heisenberg and Kac-Moody categorification
- Jun. **Northeastern** (Canada-Mexico-USA Conference in Representation Theory, Noncommutative Algebra, and Categorification): *Coulomb branches and KLRW algebras*
- May **McMaster** (Colloquium): *Representation theory and a little bit of quantum field theory*
- Apr. **Northeastern** (GASC Seminar): *Representation theory and a little bit of quantum field theory*
- Apr. **Boston U.** (geometry seminar): *Mirror symmetry for hypertoric varieties*
- Feb. **Queens** (Colloquium): *Representation theory and a little bit of quantum field theory*
- Jan. **Zoom** (Geometry and Physics Seminar, Zhejiang University): *3d mirror symmetry and symplectic singularities*
- 2021 Oct. **Zoom** (Berkeley String-Math Seminar):
Noncommutative resolutions of Coulomb branches
- Jun. **Zoom** (NSF-FRG workshop on Categorical braid group actions and categorical representation theory): *Knot homology from coherent sheaves on Coulomb branches*
- May **Zoom** (BIRS Workshop on Perspectives on Knot Homology):
Knot homology from coherent sheaves on Coulomb branches
- April **Zoom** (Kansas State M-Seminar):
Knot homology from coherent sheaves on Coulomb branches
- 2020 Nov. **Zoom** (Yale Geometry, Symmetry and Physics Seminar):
Knot homology and coherent sheaves on Coulomb branches
- Nov. **Zoom** (Northwestern Geometry and Physics Seminar): *Hypertoric mirror symmetry*
- Sep. **Zoom** (Learning Seminar on Categorification): *Coulomb branches and cylindrical KLRW algebras [2 talks]*
- Aug. **Zoom** (Quantum groups, Representation theory, Superalgebras, and Tensor categories):
Tensor products and categorification
- Aug. **Zoom** (QUACKS): *Howe to translate Gelfand-Tsetlin*
- Aug. **Zoom** (Western Hemisphere Colloquium on Geometry and Physics):
3d mirror symmetry and its discontents
- May **Zoom** (Gone Fishin'): *Coulomb branches in math and physics*
- Apr. **Zoom** (UC Davis Coulomb seminar): *Tilting bundles from positive characteristic [2 talks]*
- Feb. **Fields Institute** (Geometric Representation Theory Seminar):
Symplectic duality: where do we stand?

Professional Activities:

- Associate Chair, Graduate Studies, Waterloo Pure Math (2025 –)
- Member of Editorial Board of *Representation Theory*
- Journals refereed:

Annals of Math
Selecta Mathematica
American Journal of Mathematics
Advances in Mathematics
Representation Theory
Transactions of the AMS
Banach Center Publications
Nagoya Math Journal

Algebra and Number Theory
Journal of the EMS
Moscow Mathematical Journal
Algebraic & Geometric Topology
Quantum Topology
International Mathematics Research Notices
Duke Mathematics Journal
Mathematische Zeitschrift

Compositio Mathematica	International Journal of Mathematics
Proceedings of the LMS	& Mathematical Sciences
Journal of Algebra	Journal of the AMS
SIGMA	Inventiones Mathematicae
Fundamenta Mathematicae	Acta Mathematica
Math Annalen	Journal of the LMS
Birkhäuser Festschriften	

- External committees served:

External Review Committee for unit at Okinawa IST (2025)

AMS Committee on Publications (2015 – 2018)

AMS Web Editorial Group (2014 – 2016)

- Reviewed grants for:

NSA Mathematics

NSF Mathematics

European Research Council

BIRS workshops

EPSRC

Portuguese Foundation for Science and Technology

France Berkeley Fund

Swiss National Science Foundation

National Fund for Scientific Research (Belgium)

- Expository talks given:

Waterloo PMC

Simon's Rock College Math Club

Northeastern incoming math majors

LSU graduate colloquium

UVA Math Club

Boston Math Circle

Western Albemarle High School

Waterloo EMACS

- Conferences co-organized:

- “The geometry of double affine Hecke algebras and Coulomb branches” conference at ICMS in Edinburgh (2023)
- “QFT for Mathematicians II” summer school at Perimeter Institute (2022)
- “Perspectives on knot homology” planned for Banff; held online (2021)
- “Geometric Representation Theory” planned for Perimeter Institute and MPIM Bonn; held online (2020)
- “Algebraic and Geometric Categorification” at CMO in Oaxaca (2019)
- “QFT for Mathematicians” summer school at Perimeter Institute (2019)
- “Representation theory, mathematical physics and integrable systems” in Luminy (2018)
- “Virginia Topology Conference 2016” in Charlottesville (2016)
- “Algebraic Groups, Quantum Groups and Geometry” in Charlottesville (2016)
- “Categorification and Geometric Representation Theory” in Montréal (2014)
- “Workshop on Quiver Representations and Geometric Representation Theory” in Paris (2014)
- “Algebra, Combinatorics and Representation Theory: an international conference in memory of Andrei Zelevinsky” in Boston (2013)
- “Representation Theory and Geometry” in Berkeley (2005)

Teaching Activities:

At Waterloo:

2025 Fall Taught “Commutative Algebra” (MATH 446)

2025	Winter	Taught “Modern Algebraic Geometry” (MATH 446)
2024	Fall	Taught “Commutative Algebra” (MATH 446)
2024	Winter	Taught “Representation Theory of Finite Groups” (MATH 445)
2023	Fall	Taught “Commutative Algebra” (MATH 446)
2022	Winter	Taught “Algebraic Geometry” (MATH 464)
2021	Fall	Taught “Topics in Algebra: Category $\mathcal{O}$ ” (MATH 945)
2021	Winter	Taught “Topics in Geometry: Symplectic Geometry” (MATH 965)
2020	Fall	Taught “Linear Algebra 2 for Honours Mathematics” (MATH 235)
2020	Spring	Taught “Euclidean Geometry” (MATH 320)
2020	Winter	Taught “Calculus 1 for Honours Mathematics” (MATH 128)
2019	Fall	Taught “Linear Algebra 2 for Honours Mathematics” (MATH 235)
2019	Winter	Taught “Linear Algebra 1 for Honours Mathematics” (MATH 136)
2018	Fall	Taught “Representation Theory of Finite Groups” (PMATH 4/745)
2018	Winter	Taught “Combinatorial Representation Theory” (PMATH 945)
2017	Fall	Taught “Representation Theory of Finite Groups” (PMATH 4/745)

At UVA, NU, UO and MIT:

undergrad: Calculus (single and multivariable), Differential Equations with Linear Algebra, Transition to Higher Mathematics, Number Theory, Project Lab in Mathematics, Representations of Finite Groups

graduate: Algebra, Measure Theory, Lie Groups, Homological, Algebraic Topology, Symplectic Geometry, Quiver Representations, reading courses on categorification, geometric representation theory and quiver varieties.